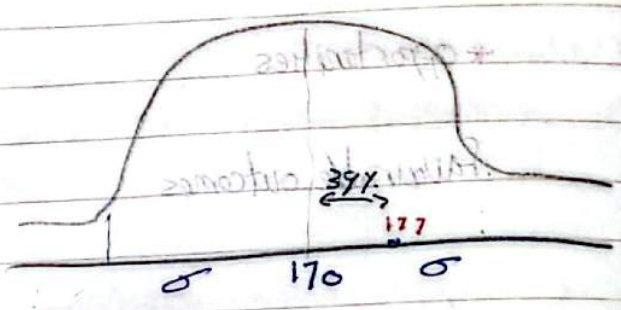
acc

$$* Z = \frac{Goal - \mu}{\sigma}$$



مثلاً لو عايز احتمال 177

$$Z = \frac{177 - 170}{5} = \boxed{0.07}$$

ارقام عشوائية للتوضيح

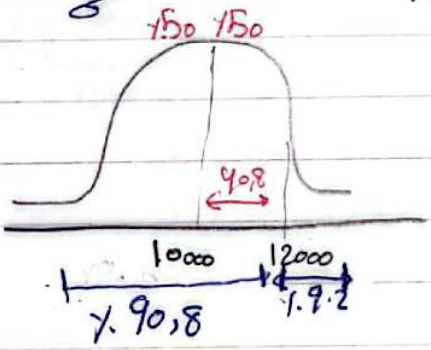
ده مش الاحتمال ده اللي هنطلعه عنه في الجدول

	0,01	0,02	0,03	0,04	0,05	0,07
0,1						
0,2						
0,3						
0,4						
0,5						
0,6						$\boxed{0.39}$ → 39%

ex $\mu = 10000$ unit breakeven = 12000 S.D = 1500

atleast ? / atmost B.e ?

$$Z = \frac{Goal - \mu}{\sigma} = \frac{12000 - 10000}{1500} = 1.33$$



	0,01	0,02	0,03
1,3			$\boxed{0.408}$ % 40,8

(ex)

2018	3500
2019	4000
2020	5000
2021	6000
2022	6500

- selling price = 10$
- V.C = 8$
- S.d = 1300
- F.C = 8000$

? → probability profit between 1000$ - 4000$

$$\mu = \frac{3500 + 4000 + 5000 + 6000 + 6500}{5} = 5000\ \$$$

$$\text{target} = \frac{F.C + \text{Profit}}{C.M}$$

C.M → pric - v.c

$$= \frac{8000 + 1000}{2} = 4500 \text{ unit}$$

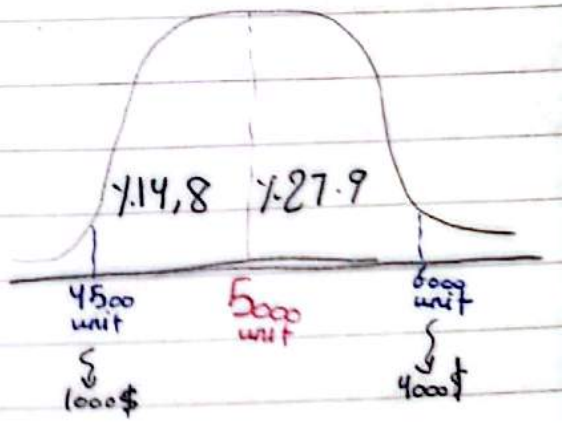
عشان نكسب 1000$ انبيع 4500 unit

$$= \frac{8000 + 4000}{2} = 6000 \text{ unit.}$$

عشان نكسب 4000$ انبيع 6000 unit

$$Z = \frac{\text{Goal} - \mu}{\sigma} = \frac{4500 - 5000}{1300} = 0{,}384$$

$$Z = \frac{6000 - 5000}{1300} = 0{,}769$$

	0,08	0,07
0,3	0,148	
0,7		0,279

* probability between 1000 - 4000 = 14.8 + 27,9
* probability more than 4000 = 50 - 27,9
* probability less than 1000$ = 50 - 14.8